

UMI-Prior: Transfer UMI Manipulation Skill to Humanoid Whole-body-Manipulation via Real-Time Motion Generation

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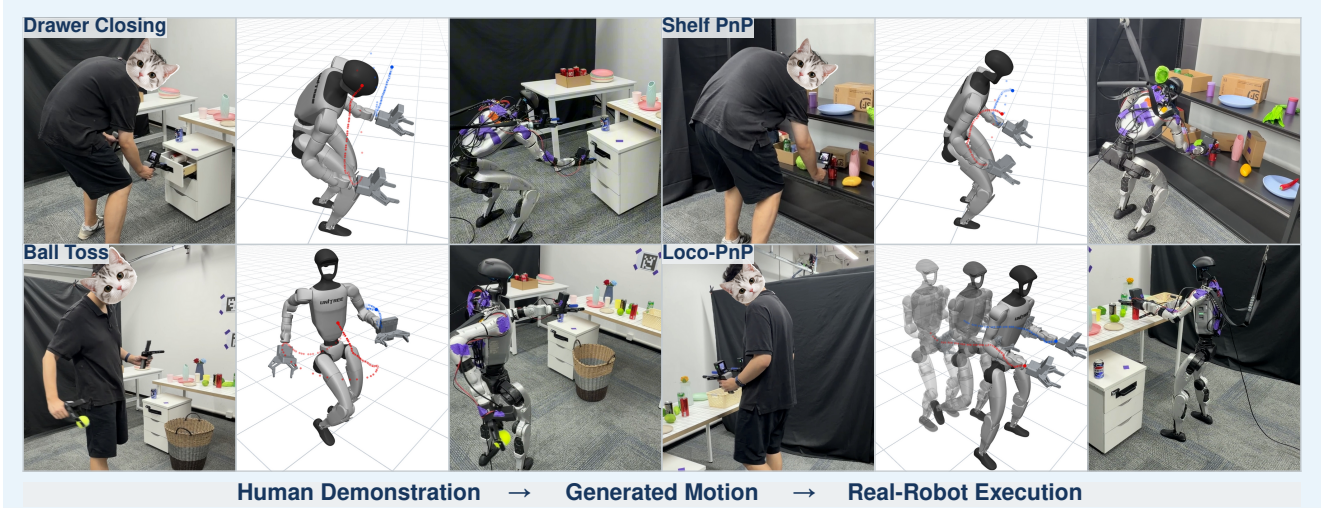


Fig. 1. **UMI-Prior** system demonstrations. Human-collected UMI trajectories are converted into whole-body motion references and executed by a real humanoid across four tasks.

Abstract—Humanoid loco-manipulation needs task-specific demonstrations and whole-body coordination, yet collecting both is costly. UMI provides end-effector trajectories but omits whole-body coordination. We present UMI-Prior, a task-agnostic generator trained on independent motion-capture data that converts native UMI trajectories into whole-body references without body trackers or paired demonstrations. On real deploy experiments, the system achieves 90% drawer-closing and 80% shelf-PnP success and qualitatively demonstrates ball toss and Loco-PnP. Together, these results show that UMI-Prior transfers UMI manipulation skills to humanoid whole-body manipulation through real-time motion generation.

I. INTRODUCTION

Which kind of data and interface, does humanoid loco-manipulation actually need to satisfy both task-centric demonstration and whole-body coordination? Teleoperation gives the most direct and reliable task-specific data, but its collection is time consuming. In contrast, UMI offers a more practical data-collection paradigm that outputs end-effector (EE) trajectories as a task-centric interface [1]. It has since been extended to quadrupeds [2] and dual-arm wheeled platforms for loco-manipulation [3]. Can we also transfer this interface to humanoid loco-manipulation?

The main problem is that an EE trajectory under-

constrains a high-DoF humanoid and cannot elicit reasonable whole-body coordination. A native solution is to add more trackers on the feet and root, extending UMI from two hand keypoints to five [4, 5], hoping to capture the whole-body coordination during the data collection. However, *simply adding body-tracking keypoints to UMI introduces new challenges without solving the core problem of learning diverse, adaptable whole-body coordination from sparse keypoints*:

- **Increased collection burden:** adding body trackers requires users to wear and calibrate multiple devices before data collection. This increases setup time and reduces the portability and convenience of the original UMI interface.
- **Limited coordination diversity:** the recorded whole-body motion reflects only how a particular demonstrator happened to perform a specific task, covering a narrow repertoire of coordination strategies. Coordination learned from such task-specific data can hardly adjust the whole-body posture flexibly as the task demands change.
- **Limited scalability of paired data:** Learning whole-body coordination from UMI demonstrations requires synchronized pairs of EE trajectories and whole-body motions. This reliance on explicit pairing prevents the effective use of large-scale

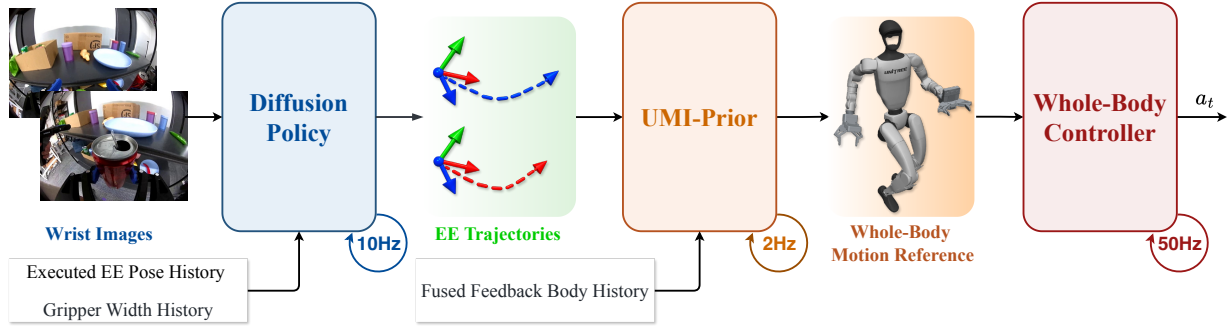


Fig. 2. Closed-loop deployment pipeline. A 10-Hz behavior-cloning policy predicts EE trajectories, **UMI-Prior** converts them into whole-body references at 1–2 Hz, and a 50-Hz WBC executes them. Measured state feeds back to both the policy and **UMI-Prior**.

unpaired motion datasets, creating a major bottleneck to scaling humanoid loco-manipulation across diverse tasks and behaviors.

Therefore, in this paper, to solve these problems, our goal is to preserve the convenience of UMI-style data collection while recovering the humanoid’s whole-body coordination with UMI origin interface, so that manipulation skills demonstrated with UMI can be naturally transferred to humanoid loco-manipulation.

Two maturing ingredients make this pipeline possible. On the data side, large-scale motion-capture data (mocap) already contain abundant whole-body coordination knowledge, and the EE trajectory can be extracted from them as the condition for free which is exactly what a conditional motion generator needs [6, 7], with no paired collection ever performed. On the execution side, the whole-body controller (WBC) is increasingly mature [8, 9], exposing a standard interface: a kinematic whole-body reference. The problem therefore becomes interface translation: how can UMI’s EE interface be converted into a whole-body reference for a WBC? We address this question with our hierarchical loco-manipulation pipeline (Fig. 2), built around **UMI-Prior**, a task-agnostic motion generator trained independently on existing mocap data. A behavior-cloning (BC) policy produces EE trajectories. Conditioned on these trajectories, **UMI-Prior** synthesizes coordinated whole-body references, which are executed by a WBC. Representative **UMI-Prior** generations are shown in Fig. 3.

In summary, our contributions include:

- 1) **A general, plug-in UMI-to-humanoid interface.** It retains native EE-only UMI demonstrations, requiring no body trackers, whole-body motion capture, or pelvis and foot localization. The independently trained, task-agnostic **UMI-Prior** completes the missing whole-body motion through

an explicit, inspectable reference layer. Its EE and whole-body interfaces decouple the BC policy from the WBC and support compatible policies and controllers.

- 2) **UMI skill transfer to humanoid whole-body manipulation.** Trained on independent mocap, **UMI-Prior** generates coordinated whole-body references in real time while preserving the commanded EE motion. This transfers skills from UMI’s EE space to humanoid loco-manipulation without paired UMI and whole-body demonstrations.
- 3) **A closed-loop three-layer deployment system.** The BC policy predicts EE trajectories, **UMI-Prior** converts them into whole-body references in real time, and the WBC executes them. Executed-state feedback couples all three layers at each receding-horizon replanning cycle.

II. RELATED WORK

Our pipeline needs three ingredients: UMI-style portable data collection, conditioned motion generation, and whole-body control. Each is mature on its own, but they have not been connected for the humanoid.

A. UMI and the Humanoid Whole-Body Gap

UMI enables portable, robot-free collection of manipulation demonstrations and provides EE trajectories as a task-centric interface [1, 10]. Recent extensions apply this interface to legged and dual-arm mobile platforms [2, 3]. They show that EE trajectories transfer manipulation intent across embodiments, but how to effectively leverage them for humanoid whole-body control remains an open problem.

A humanoid is a high-DoF, floating-base system, so EE trajectories underdetermine its whole-body motion and cannot enforce both task execution and coordination. Conventional inverse kinematics and trajectory

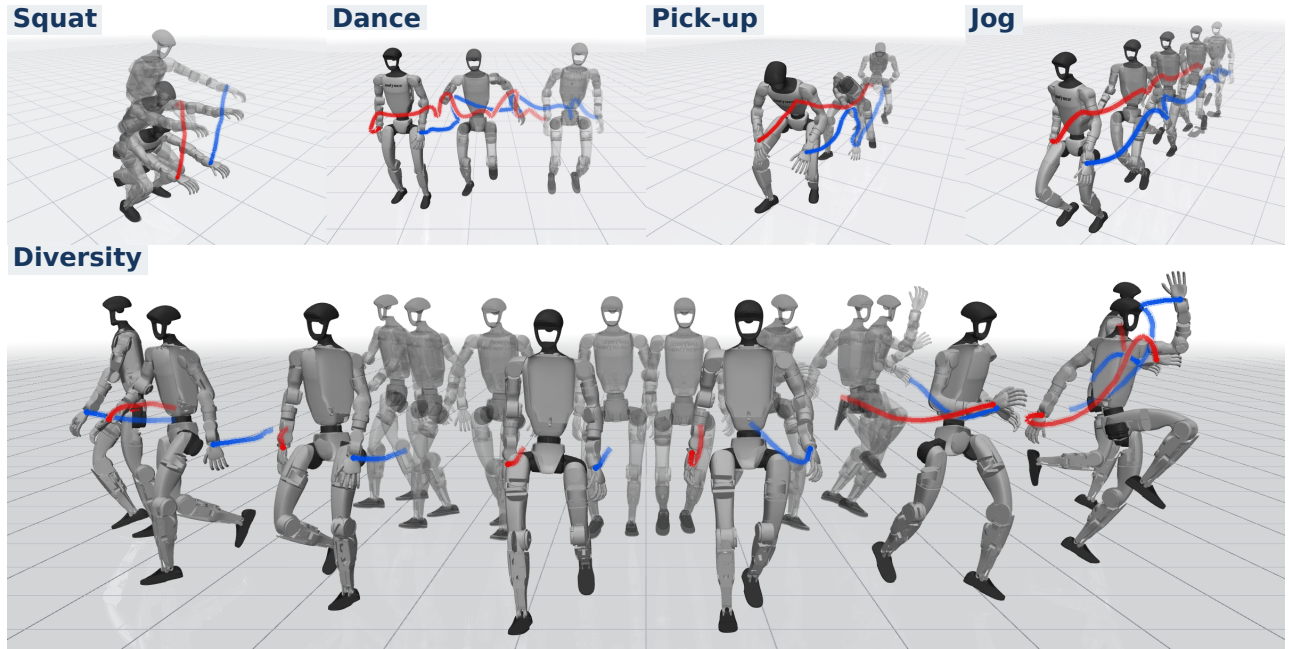


Fig. 3. Qualitative **UMI-Prior** generations. Top: whole-body completions for squat, dance, pick-up, and jog. Bottom: diverse completions conditioned on the dual-hand EE paths shown in red and blue.

optimization do not resolve this ambiguity [11, 12].

Recent humanoid systems address this gap during collection by recording full-body human motion or sparse body keypoints from wearable VR trackers for retargeting [4, 5]. These measurements expose coordination but increase collection overhead and require each EE trajectory to be paired with body motion. We retain the native UMI interface and instead train a task-agnostic motion prior on independent motion data. At runtime, it generates coordinated whole-body motion that follows the commanded EE trajectory rather than reconstructing the demonstrator’s unobserved body motion.

B. Conditioned motion generation

Motion diffusion models learn distributions of plausible whole-body motion and expose them as controllable priors for text, sparse observations, and kinematic constraints [6, 7, 13]. This makes them suitable for resolving the ambiguity left by an EE interface. Large-scale motion data are also becoming available [14]. Since EE conditions can be extracted from each sequence through forward kinematics, a generator can learn from existing EE-whole-body pairs. This removes separately paired UMI and body recordings and keeps training independent of the downstream manipulation policy.

Online deployment further requires causal, low-latency generation. Autoregressive diffusion models plan from recent history and streaming text or kinematic constraints in real time [16, 17]. They also support

EE control and humanoid tracking [17], but serve as general-purpose interactive motion generators. Our setting instead requires a lightweight planner that preserves short-horizon EE commands while producing a whole-body reference at each replanning step. Table I compares this interface with the closest controllable motion generators.

C. Whole-body controller

Motion-tracking controllers execute diverse whole-body references and maintain tracking under disturbances [8, 9, 18]. They expose a motion-reference interface that maps desired trajectories to low-level robot actions, allowing reference generation and control to be developed separately.

Behavior foundation models instead place multiple skills in one low-level policy. They encode behaviors as learned latent variables and train on large motion datasets [19, 20]. A latent variable selects the behavior, while the policy directly produces robot actions. This supports diverse skills, but keeps the motion prior implicit, complicates implementation, and provides no separate reference interface.

We instead retain a versatile motion tracker downstream and implement **UMI-Prior** as a separate reference-generation layer. Its output is explicit and inspectable, while the tracker can be replaced without retraining the motion prior. This separation keeps behavior modeling outside the low-level control stack.

TABLE I
Capabilities of controllable motion generators.

Method	Online generation		Spatial control		Native control		Robot loop		Context (s)	
	Real-time	Online	EE pos.	EE rot.	No opt.	No RL	Hardware	Feedback	Hist.	Future
MaskControl [15]	×	×	✓	×	×	✓	×	×	—	10.00
Kimodo [13]	×	×	✓	✓	✓	✓	×	×	—	10.00
DartControl [16]	✓	✓	✓	×	×	×	×	×	0.07	0.27
ARDY [17]	✓	✓	✓	✓	✓	✓	✓	×	8.00	10.00
UMI-Prior (ours)	✓	✓	✓	✓	✓	✓	✓	✓	0.27	2.13

III. METHOD

Table II defines the sequence notation used throughout this section.

TABLE II
Sequence notation used by the policy and **UMI-Prior**.

Notation	Description
—	History observation, $1 \leq i \leq H$
+	Future segment, $H < i \leq H + F$
⊙	Elementwise multiplication
i	Frame index
F/H	Predicted-future (32) / history (8) horizons
T_{DP}	Diffusion-policy prediction horizon
Sample_d	Every d -th frame from a to b , with temporal stride d : $\text{Sample}_d((\mathbf{x}_i)_{i=a}^b) = (\mathbf{x}_{a+kd})_{k=0}^{\lfloor (b-a)/d \rfloor}$.

A. Problem Formulation and System Overview

As shown in Fig. 2, our system contains three modules: a BC policy implemented as a diffusion policy (Table III), **UMI-Prior** (Table V), and a WBC. First, the diffusion policy takes wrist RGB observations, an EE trajectory history, and a gripper-width history as input. It predicts the EE condition $\mathbf{c}$ together with gripper-width commands, where $\mathbf{c} = (\mathbf{S}^{ee}, \mathbf{L}^{ee})$.

$\mathbf{S}^{ee} = (\mathbf{s}_i^{ee})_{i=1}^{H+F}$, $\mathbf{L}^{ee} = \text{Sample}_d((\mathbf{s}_i^{ee})_{i=H+1}^{T_{DP}})$. (1)
 $\mathbf{S}^{ee}$ contains the EE history and frame-aligned constraints for the current F -frame prediction. The look-ahead $\mathbf{L}^{ee}$ previews later DP commands as future intent rather than frame-wise targets. It helps **UMI-Prior** coordinate the body beyond the current window without forcing the generated motion to follow the less certain tail of the receding-horizon plan. Each $\mathbf{s}_i^{ee}$ follows the EE representation in Table IV.

TABLE III
Behavior-cloning policy configuration.

Hyperparameter	Value
Observation horizons (visual / proprioceptive)	2 / 3
Observation / action frequency	10 / 10 Hz
Action horizon	40
Image resolution (two cameras)	$2 \times 224 \times 224$
Vision backbone	DiNOv3 ViT-B/16
Policy backbone	DiT; 6 blocks, width 384, 6 heads
Learning rates (policy / vision)	$10^{-4} / 10^{-5}$
Epochs / batch size	100 / 16
Training loss	Flow matching
Inference steps	8

Second, **UMI-Prior** which will be introduced later, conditions on the body history $\mathbf{S}^{\text{body},-}$ and $\mathbf{c}$ to predict

the whole-body future $\mathbf{S}^{\text{body},+}$, where the whole-body sequence is

$$\mathbf{S}^{\text{body}} = (\mathbf{s}_i^{\text{body}})_{i=1}^{H+F} = [\mathbf{S}^{\text{body},-}, \mathbf{S}^{\text{body},+}]. \quad (2)$$

Each $\mathbf{s}_i^{\text{body}}$ follows the whole-body representation in Table IV. Third, the pretrained SONIC motion tracker [9] serves as the WBC and tracks the generated reference.

B. UMI-Prior: Real-Time EE-Conditioned Generation

Following TextOp [21], we model long-horizon motion as a sequence of short, fixed-length motion primitives. A motion primitive is an F -frame whole-body trajectory segment generated from the observed body history and the corresponding EE condition $\mathbf{c}$. Figure 4 summarizes the training and inference framework. **UMI-Prior** learns this single-primitive mapping and applies the same generator twice to produce a $2F$ -frame rollout. For training, we re-encode retargeted mocap using the robot-skeleton representations in Table IV and extract $\mathbf{c}$ from its wrist trajectories.

TABLE IV
Representations of the whole-body state and EE condition.

Component	Representation	Size
Whole-body state $\mathbf{s}_i^{\text{body}}$		Sum: 73
Root orientation	Roll/pitch sine-cosine; yaw increment	5
Root position	Translation increment, height, planar pose	8
Joint motion	Joint positions and increments	58
Foot contact	Left/right contact	2
EE condition $\mathbf{s}_i^{ee}$		Sum: 50
Relative pose	Segment-relative position and 6D orientation	18
Height anchor	Initial heights	2
Orientation anchor	Initial heading-frame 6D orientations	12
Pose increment	Position and orientation increments	18

Single-primitive generation. We train **UMI-Prior** with masked conditional flow matching. Let ϵ be Gaussian noise matching $\mathbf{S}^{\text{body}}$. The mask $\mathbf{m}$ fixes history frames and marks future frames for generation:

$$m_i = \begin{cases} 1, & 1 \leq i \leq H, \\ 0, & H < i \leq H + F. \end{cases} \quad (3)$$

The flow time $\tau \in [0, 1]$ controls the future interpolation from noise at $\tau = 0$ to ground-truth motion at $\tau = 1$, while the history remains fixed:

$$\tilde{\mathbf{z}}_\tau = \mathbf{m} \odot \mathbf{S}^{\text{body}} + (1 - \mathbf{m}) \odot [(1 - \tau)\epsilon + \tau \mathbf{S}^{\text{body}}]. \quad (4)$$

We use a DiT backbone parameterized by θ , with each whole-body frame embedded as a body token. The

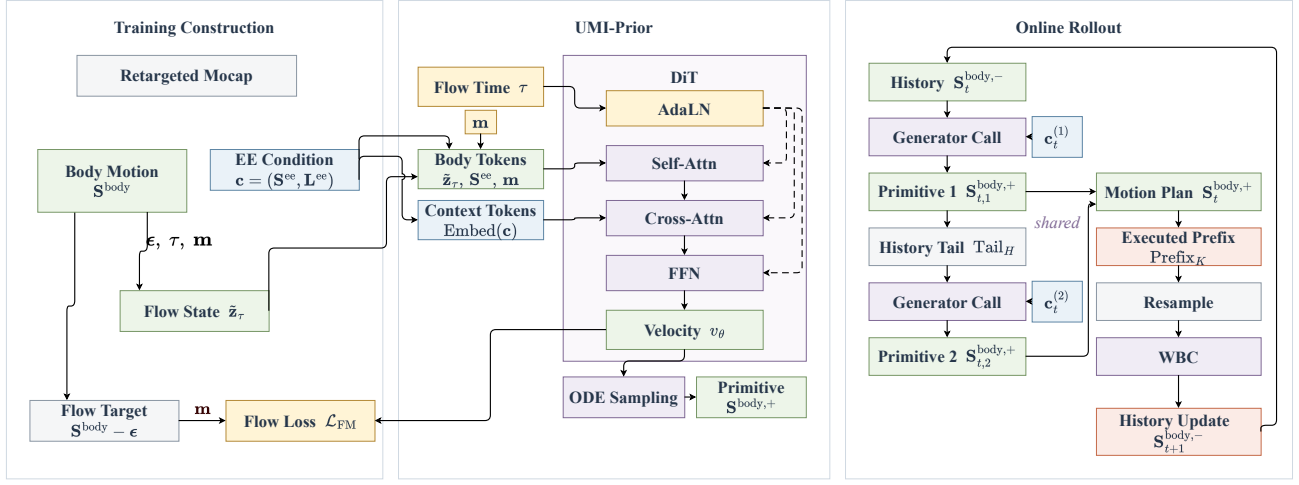


Fig. 4. **UMI-Prior** training and online generation. Retargeted mocap provides EE conditions and whole-body targets for training, while policy-predicted EE commands and measured history drive two-primitive rollout and feedback-based replanning.

model adds $\mathbf{S}^{\text{ee}}$ to the time-aligned tokens, while all body tokens attend to $\mathbf{c}$ through cross-attention. Given $\tilde{\mathbf{z}}_\tau$ and τ , the conditioned DiT predicts the flow velocity $v_\theta(\tilde{\mathbf{z}}_\tau, \tau, \mathbf{c})$. We supervise this prediction only on future frames:

$$\mathcal{L}_{\text{FM}} = \mathbb{E} \left\| (1 - \mathbf{m}) \odot \left[v_\theta(\tilde{\mathbf{z}}_\tau, \tau, \mathbf{c}) - \mathbf{S}^{\text{body}} + \epsilon \right] \right\|_2^2. \quad (5)$$

For each primitive, we further apply $\mathcal{L}_{\text{state}}$, $\mathcal{L}_{\text{EE}}$, and $\mathcal{L}_{\text{coord}}$ for state accuracy, EE tracking, and whole-body coordination, respectively.

Two-primitive rollout. Following TextOp [21], we apply the same trained generator twice to form a $2F$ -frame reference. The first primitive uses the observed history $\mathbf{S}_t^{\text{body},-}$ and its corresponding EE condition. The end of the first prediction becomes the input history of the second primitive, which uses the subsequent EE condition. Concatenating the two predicted future segments gives the whole-body reference.

During training, we gradually replace the second primitive’s ground-truth history with generated history and perturb the input histories. We supervise the stitched prediction and use $\mathcal{L}_{\text{seam}}$ to enforce continuity between the two primitives. Combining the single-primitive and rollout objectives with nonnegative weights $\lambda_s, \lambda_e, \lambda_c, \lambda_b$ gives

$$\mathcal{L} = \mathcal{L}_{\text{FM}} + \lambda_s \mathcal{L}_{\text{state}} + \lambda_e \mathcal{L}_{\text{EE}} + \lambda_c \mathcal{L}_{\text{coord}} + \lambda_b \mathcal{L}_{\text{seam}}. \quad (6)$$

C. Integrated Deployment System

At runtime, the behavior-cloning policy runs at 10 Hz. It uses the latest wrist observations, executed EE history, and gripper widths to predict an EE trajectory. We

TABLE V
UMI-Prior architecture and training configuration.

Hyperparameter	Value
History / prediction horizon	8 / 32
Number of primitives	2
Motion frequency	30 Hz
EE look-ahead horizon / stride	32 / 4
Temporal backbone	DiT; 10 blocks, width 384, 6 heads
Learning rate	5.66×10^{-4}
Epochs / batch size	300 / 8192
Training loss	Masked flow matching
Inference ODE steps	16

convert this trajectory to $\mathbf{c}$ and resample it to 30 Hz for **UMI-Prior**. Gripper commands follow the same timing but bypass **UMI-Prior**. Given $\mathbf{c}$ and the current whole-body history, **UMI-Prior** generates two primitives, yielding 64 future frames.

We use receding-horizon execution. While the WBC tracks the current reference, **UMI-Prior** generates the next one. The system commits the first 32 planner frames. The remaining frames serve as a preview and are replaced by the next accepted plan. We resample the reference from 30 to 50 Hz for the WBC. This gives a 1.875-Hz replanning rate while the WBC continues at 50 Hz.

At each boundary, we align controller feedback with the executed reference and resample it to 30 Hz. Measured joint states and base orientation update the next whole-body history. The same feedback updates the EE history for the next policy query. This loop corrects tracking errors while preserving future EE commands.

IV. EXPERIMENTS

Our experiments address three questions that correspond to the paper’s contributions:

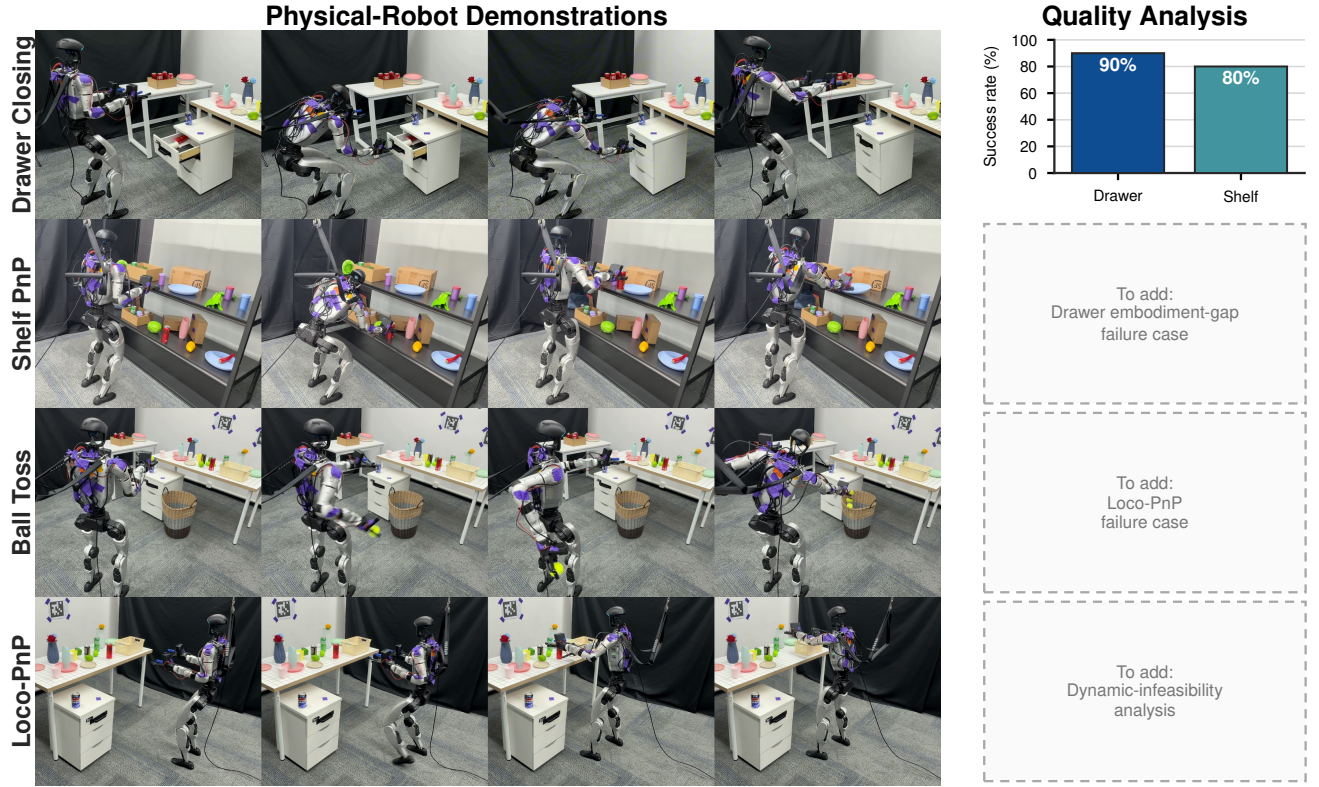


Fig. 5. Real-robot transfer with **UMI-Prior**. Left: drawer closing, shelf PnP, ball toss, and Loco-PnP. Right: success rates for the two quantitatively evaluated tasks and three reserved analyses: the drawer embodiment gap, the Loco-PnP failure case, and dynamic infeasibility.

TABLE VI
Quantitative analysis of **UMI-Prior**.

Method	EE pos. (cm)↓	EE vel. (m/s)↓	EE orient. (deg)↓	Foot vel. (m/s)↓	Joint rot. (deg)↓	Joint pos. (m)↓	Limit viol. (%)↓	FID ↓
Recorded motion (reference)	-	-	-	0.008	-	-	-	-
EE+root (oracle)	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
EE+root+feet (oracle)	3.87	0.0323	1.78	0.0236	2.56	0.0163	15.22	0.0079
UMI-Prior w/o look-ahead	5.77	0.0438	2.52	0.0590	5.31	0.0583	13.16	0.0280
UMI-Prior	4.25	0.0364	1.42	0.0536	4.27	0.0439	13.47	0.0202

- Q1: Reference completion.** Can **UMI-Prior** complete EE-only UMI trajectories into coordinated whole-body motion, and how do look-ahead and the absence of root and foot observations affect completion quality?
- Q2: Real-time execution in simulation.** Can **UMI-Prior** generate references in real time for execution, and does measured-state feedback improve execution?
- Q3: Real-world transfer.** Can the complete hierarchy transfer native UMI policies to successful humanoid whole-body-manipulation?

A. Reference Completion

Setup. We evaluate **UMI-Prior** offline, without WBC execution, to isolate its ability to convert EE trajectories into whole-body references. We compare the full model

with a no-look-ahead ablation that uses the same data, backbone, and sampling procedure but does not receive L^{ee} . To quantify the gap caused by missing body measurements, we also evaluate analysis-only oracle conditions given the ground-truth root pose or both root and foot poses. These signals are unavailable in native UMI demonstrations, so their gap to the EE-only model measures the cost of tracker-free data collection.

We build the G1 motion dataset from BONES-SEED [14]. We retain basic locomotion and manipulation motions, together with simple exercises and dances that provide dynamic whole-body motion, while excluding complex motions that rarely occur in loco-manipulation. The dataset contains about 105 hours of motion: 90 hours for training, 5 hours for validation, and 10 hours for testing. We split clips by semantic action

group, with no group shared across splits; therefore, the test categories are unseen during training.

Metrics. Here, we introduce the motion-quality metrics used in Table VI. For each metric, we first average over frames within each test clip and then report the mean across clips. EE position, velocity, and orientation measure tracking errors to the commanded EE trajectory. Foot velocity measures support-foot speed during contact. Joint rotation and position measure angular and forward-kinematic position errors against the recorded whole-body motion, while limit violation is the percentage of frames with any joint outside its range.

FID assesses the distributional similarity between generated and real motions in the extracted feature space; for their empirical feature distributions P_g and P_r , with means (μ_g, μ_r) and covariances (Σ_g, Σ_r) , it is computed as

$$\text{FID} = \|\mu_g - \mu_r\|_2^2 + \text{Tr}\left(\Sigma_g + \Sigma_r - 2(\Sigma_g \Sigma_r)^{1/2}\right), \quad (7)$$

where lower values indicate closer motion distributions.

Results. Figure 3 provides a qualitative assessment of the generated motions and their diversity. The top row shows coherent whole-body motions generated by **UMI-Prior** for squat, dance, pick-up, and jog, covering whole-body coordination, dynamic motion, manipulation, and locomotion. For the diversity test, we use multiple hand trajectories with different speeds as conditions. The bottom row shows distinct whole-body motions that follow the conditioned EE paths. Table VI quantitatively shows that EE look-ahead reduces most tracking errors and FID. Oracle root and foot inputs further improve whole-body coordination and reduce foot sliding, revealing the cost of the tracker-free EE-only setting. They also reduce ambiguity in body placement, preventing overextended arm postures and encouraging more coordinated torso and leg motion.

B. Real-Time Execution in Simulation

Setup. Following the offline evaluation, we test whether **UMI-Prior** can plan executable whole-body references in real time for simulated shelf PnP. We use a pretrained low-latency SONIC controller as the WBC and bypass the diffusion policy by supplying ground-truth EE commands. Both **UMI-Prior** conditions are evaluated over 100 rollouts. We compare **UMI-Prior** with **UMI-Prior+FB**, which uses the measured robot state at each replanning step, to test the effect of feedback. As an EE-only baseline, we separately train a WBC that directly receives the EE commands, testing

whether direct EE control is sufficient without **UMI-Prior**. The intended confirmatory comparison shares commands, initial states, and safety limits. The EE-only result below is instead a preliminary single full-scale shelf rollout; the confirmatory multi-rollout evaluation remains to be completed.

TABLE VII

Simulation tracking errors for shelf PnP. Lower is better. Bold marks the numerically lowest available value in each row. The EE-only entries report one preliminary nominal-only full-scale shelf rollout and are not directly comparable with the mixed-condition **UMI-Prior** aggregates.

Metric ($\downarrow$)	EE only	UMI-Prior	UMI-Prior+FB
UMI-Prior	×	✓	✓
Feedback	×	×	✓
EE vel. (m/s)	0.144	0.078	0.084
EE orient. (deg)	30.00	6.76	4.97
Foot slip (m/s)	0.002	0.009	0.010
Root-rel. MPJPE (m)	-	0.024	0.017
Joint rot. err. (rad)	-	0.076	0.046

Results. Root-relative MPJPE measures joint-position error after removing global translation by expressing each joint relative to the root link:

$$E_{\text{MPJPE}} = \frac{1}{TJ} \sum_{t=1}^T \sum_{j=1}^J \left\| (\mathbf{p}_t^{\text{pol}}[j] - \mathbf{p}_t^{\text{pol}}[0]) - (\mathbf{p}_t^{\text{ref}}[j] - \mathbf{p}_t^{\text{ref}}[0]) \right\|_2, \quad (8)$$

where $\mathbf{p}^{\text{pol}}$ and $\mathbf{p}^{\text{ref}}$ denote the executed and reference joint positions, respectively, and index 0 denotes the root. Joint-rotation error is likewise an error metric: the mean absolute angular difference between the reference and executed configurations over all scoring frames and all 29 actuated joints. The EE-only WBC has no full-body reference configuration, so neither this metric nor root-relative MPJPE is applicable. Foot slip is the mean horizontal speed of both ankle links during inferred stance.

Table VII shows that measured-state feedback reduces EE orientation error, root-relative MPJPE, and joint-rotation error. EE velocity error and foot slip are slightly higher. Because failed rollouts are excluded, the no-feedback column is a selected subset (27 rollouts: 14 nominal and 13 perturbed) whereas the feedback column contains all 100 rollouts (50 nominal and 50 perturbed). Here nominal denotes no external perturbation and perturbed denotes the protocol-specified disturbance. The small reversals should therefore not be interpreted as a causal degradation from feedback. Although the single nominal EE-only rollout has a numerically lower foot-slip proxy, it is not comparable with these mixed-condition aggregates. It also has higher EE velocity and orientation errors. A controlled multi-rollout comparison remains pending.

C. Real-World Transfer

We deploy the complete hierarchy on four physical-robot tasks using the 29-DoF G1 humanoid, as shown in Fig. 5. For each task, we collect 200 demonstrations with the GenRobot DAS Gripper [22] in approximately 90 minutes. The G1 uses the corresponding DAS Controller [23], maintaining a consistent gripper interface from data collection to deployment.

Drawer Closing. The robot follows the EE plan to reach the handle, lower its body, and close the drawer while maintaining balance. The complete hierarchy achieves a 90% success rate over 10 independent trials, showing that native UMI commands can be converted into coordinated whole-body behavior. We also observe an embodiment-gap failure mode. Human demonstrators have a longer arm reach and can close the drawer by crouching, whereas G1 must step forward or lean farther to reach the same target. Although these adaptations are physically reasonable, they increase the risk of task failure.

Shelf PnP. The robot transfers a bottle from the lower shelf to the upper shelf by coordinating manipulation with whole-body motion. The complete hierarchy achieves an 80% success rate over 10 independent trials, demonstrating whole-body manipulation across different working heights.

Ball Toss. Ball toss tests whether the hierarchy can generate highly dynamic whole-body motion. The demonstrated executions show whole-body coordination with a timed release. We report qualitative results only because throwing accuracy is affected by other factors, such as gripper release latency and high-speed EE prediction, and requires substantial engineering tuning beyond the scope of this work.

Loco-PnP. The robot walks forward to reach the table and then performs pick-and-place. Because the current success rate is low, we report only qualitative results and omit the success rate. The main failure mode is unstable locomotion: **UMI-Prior** can generate kinematically feasible but dynamically infeasible references that fall outside SONIC’s training distribution. Future work will improve physical grounding through WBC data augmentation or a residual reinforcement-learning policy.

Summary. Back to the three questions. **Q1:** Yes. **UMI-Prior** completes native EE-only UMI trajectories into coordinated whole-body references, and EE look-ahead improves motion fidelity. The gap to the oracle settings measures the performance loss caused by missing pelvis and foot observations. **Q2:** Yes, in

our tests. The three-layer system plans fast enough for online execution and replans from measured robot states. However, residual EE tracking errors and foot slip still reduce reliability in dynamic motions. **Q3:** Yes for the two scored real-robot tasks. The hierarchy achieves object-level success on drawer closing and shelf PnP, while ball toss and Loco-PnP provide qualitative evidence of broader loco-manipulation transfer. Overall, the results show that our **UMI-Prior**-based hierarchy system successfully transfers UMI manipulation skills to humanoid whole-body manipulation though it still leaves limitations which will be addressed in the future work.

V. CONCLUSION

We presented **UMI-Prior**, a task-agnostic generator that converts native bimanual UMI trajectories into coordinated whole-body references learned from independent motion data. The resulting hierarchy supports real-time closed-loop execution without body trackers or paired UMI and whole-body demonstrations. Experiments show that EE look-ahead improves motion completion, simulation validates online executability, and physical-robot trials demonstrate transfer across four loco-manipulation tasks. Future work will improve physical grounding through WBC data augmentation or residual reinforcement learning, adding dynamics and contact constraints to the generated references. We will also combine large-scale UMI data with whole-body motion generation to pretrain an end-to-end model that jointly learns task intent and whole-body coordination.

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